

ABSTRACT

FocusFlow is a full-stack, client-rendered productivity web application built on Next.js 14 (App Router), React 18, TypeScript, and Tailwind CSS. The system provides an integrated suite of five productivity modules — Dashboard, Task Management, Deep Work Timer, Habit Tracker, and Notes — complemented by a passive Analytics module. The distinguishing architectural principle of FocusFlow is its Recovery-First, Emotionally Intelligent (EQ) design philosophy, which explicitly prioritises the psychological wellbeing of the user over raw output metrics.

Unlike conventional productivity tools that measure success solely through completion rates and streaks, FocusFlow incorporates an awareness of user energy states and introduces a dedicated Recovery Mode. When the system determines that the user has been unproductive for an extended period, the interface reduces cognitive load by auto-collapsing the sidebar, limiting task visibility to a maximum of three items, hiding analytics, and surfacing compassionate, non-judgmental messaging. This adaptive interface behaviour is governed by a custom `useRecoveryMode` hook that evaluates the `lastProductiveAt` timestamp stored in the Zustand-powered `userStore`.

The application architecture follows a modular, component-driven pattern. Each functional area is represented as a Next.js route containing a server-side page wrapper and a client-side Content component. Global UI state, task data, habit records, deep-work session history, and note content are managed independently through dedicated Zustand stores, enabling isolated, testable state slices without prop-drilling. The design system is built around a set of CSS custom properties (design tokens) that support both a light and dark theme via Tailwind's dark-mode class strategy.

Key features include a Conversational Reset widget on the dashboard that performs rule-based sentiment analysis on free-text user input and generates context-aware productivity suggestions; a Pomodoro-style Circular Focus Timer with gradient SVG arcs and session history logging; a block-based note editor supporting text, heading, checklist, quote, divider, and voice block types with drag-and-drop reordering; a Habit Tracker with a heatmap calendar, weekly strength scoring, and hydration ring visualisation; and a Quiet Insights analytics panel comprising a Focus Pattern Bar Chart, Habit Strength Radar, Mood Summary, and AI-style Narrative Insights.

The project was developed across four incremental phases, culminating in a locally-deployable, fully-functional application. State persistence is handled via `localStorage` through Zustand's `persist` middleware, ensuring data survival across browser sessions without requiring a backend database for the current release phase.

Keywords: *Productivity Application, Next.js, React, TypeScript, Zustand, Recovery Mode, Habit Tracker, Deep Work, Block-Based Notes, EQ-First Design, Tailwind CSS, Web Application.*